Series — 5.2

If f takes real #s as input, then

$\int_a^b f(x)dx$ "adds up" all infinitely many values of f from $f(a)$ to $f(b)$, "weighted" according to the "infinitely small" dx.

With a sequence, we only have one output for each natural #, and the analogue of integration is to add all these infinitely many values together.

Given a sequence (a_n), we get a <u>series</u> $\sum_{n=1}^{\infty} a_n$.

<u>Definition</u>: A series $\sum_{n=1}^{\infty} a_n$ is the limit of the sequence of partial sums (s_k) where

$$s_k = \sum_{n=1}^{k} a_n . \quad \text{That is, } \sum_{n=1}^{\infty} a_n = \lim_{k\to\infty} s_k .$$

$$4$$

$$a_1 + a_2 + \cdots + a_k$$

If this limit exists, we say the series $\sum_{n=1}^{\infty} a_n$ <u>converges</u> and equals this limit. Otherwise, we say that $\sum_{n=1}^{\infty} a_n$ <u>diverges</u>.

We've seen sequence before that are partial sums:

- $(1, 2, 3, 4, \ldots)$ is the sequence of partial sums of the sequence $(1, 1, 1, \ldots)$

- $(3, 3.1, 3.14, 3.141, \ldots)$ is the sequence of partial sums of the sequence $(3, \frac{1}{10}, \frac{4}{100}, \frac{1}{1000}, \ldots)$

Series do not have to start with $n=1$

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} = \sum_{n=5}^{\infty} \frac{1}{2^{n-4}}$$

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

<u>Example</u>: What does $\sum_{n=1}^{\infty} \frac{1}{2^n}$ equal? Does it converge?

Taking the limit of the sequence

$$\left(\frac{1}{2}, \frac{1}{2}+\frac{1}{4}, \frac{1}{2}+\frac{1}{4}+\frac{1}{8}, \frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}, \ldots\right)$$ of partial sums

$$= \left(\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \ldots\right). \quad \therefore \sum_{n=1}^{k} \frac{1}{2^n} = \frac{2^k-1}{2^k} = 1 - \frac{1}{2^k}$$

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \lim_{k\to\infty} \sum_{n=1}^{k} \frac{1}{2^n} = \lim_{k\to\infty} 1 - \frac{1}{2^k} = 1$$ — convergent!

Example: Does $\sum_{n=1}^{\infty} (-1)^n$ converge or diverge?

The partial sums are $(-1, 0, -1, 0, -1, 0, \dots)$ which does not converge, so $\sum_{n=1}^{\infty} (-1)^n$ diverges.

Properties: $\sum_{n=1}^{\infty} c a_n = c \sum_{n=1}^{\infty} a_n$, when $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent, we also get

$$\sum_{n=1}^{\infty} a_n \pm b_n = \sum_{n=1}^{\infty} a_n \pm \sum_{n=1}^{\infty} b_n .$$

There is a multiplication formula, but it's a bit difficult to use.